1.PROGRAM DESIGN

NAME OF ACTIVITY	LEMS: Library Entrance Monitoring System
IMPLEMENTATION DATE	February 2026 – April 2026
VENUE	Learning Information and Resource Center Cor Jesu College, Inc., Digos City
PARTICIPANTS	Rhea Grace P. Balatero John Mark P. Limsan Library Staff and Administrators

I. RATIONALE

The Library Office currently tracks users manually through logbooks, a process that is time-consuming, prone to errors, and limited in its ability to produce accurate reports. As the student population continues to grow, it becomes increasingly important to monitor library visitors efficiently, particularly in managing peak hours, ensuring security and access tracking, generating reliable usage reports, and improving overall operational efficiency. To address these needs, the **Library Entrance Monitoring System (LEMS)** aims to **digitize and automate the entire entrance monitoring process, making it faster, more accurate, and more dependable for both staff and administrators.**

Beyond improving library operations, the implementation of LEMS also aligns with several United Nations Sustainable Development Goals (SDGs). By enhancing the quality, orderliness, and accessibility of library services, the project supports **SDG 4: Quality Education**, ensuring that students benefit from a well-managed and technology-enabled learning environment. Through its innovative digital approach, LEMS contributes to **SDG 9: Industry, Innovation, and Infrastructure** by introducing modern technological solutions that strengthen institutional systems and promote a culture of digital advancement. Furthermore, by enhancing transparency, accountability, and security in tracking library access, LEMS supports **SDG 16: Peace, Justice, and Strong Institutions**, contributing to a more efficient and trustworthy institutional process.

II. OBJECTIVES

1. To design and develop an automated Library Entrance Monitoring System that improves efficiency, accuracy, and security in tracking library users.
2. To digitally capture entrance and exit logs of students and visitors.
3. To monitor daily library attendance, including peak hours and user categories.
4. To generate automated reports for administrative use.
5. To enhance security by providing verified and timestamped entry records.
6. To provide a user-friendly interface for both library staff and administrators.

III. TECHNIQUES

1. The proponents (Rhea Grace P. Balatero and John Mark P. Limsan) will coordinate closely with the Library Office and the Software Engineering Instructor (Mdm. Ciemavil S. Alcain) to ensure alignment with institutional needs and project requirements.
2. The team will gather initial data through interviews, observations, and document reviews to design an appropriate system model.
3. Development activities will utilize collaborative tools such as Google Workspace for documentation, planning, version control, and communication.
4. Regular consultations with the Library Staff will be conducted to validate requirements and ensure continuous feedback throughout the development process.
5. The Multimedia Team will assist in creating wireframes, interface mockups, and system demonstration materials.

IV. METHODOLOGY / PROGRAM FLOW

- LEMS will follow a phased development approach:
  - Phase 1: Data Gathering & Analysis (February 2026)
    - Conduct interviews and process observations
    - Gather operational data and identify system requirements
  - Phase 2: System Design & Development (March 2026)
    - Draft SRS, ERD, DFDs, and UI mockups
    - Begin coding, database creation, and interface development
  - Phase 3: Testing, Deployment & Evaluation (April – May 2026)
    - Conduct system testing and debugging
    - Finalize deployment version
    - Present the system to the Library Office and gather feedback

V. PLANNING AND ORGANIZING

DATE	OBJECTIVES	ACTIVITY	MATERIALS NEEDED	PERSON-IN-CHARGE
February 3 - 13, 2026	Begin project preparation	Conduct initial interviews; Gather library operational data	Interview guide, notebook, laptop, cellphone	Proponents
February 14 - 22, 2026	Analyze requirements	Draft SRS and system models	Google Docs, diagrams	Proponents

February 23 - March 15, 2026	Develop system components	Coding, database creation, UI design	Laptop, IDE, MySQL	Proponents
March 16 - 20, 2026	Validate design	Present prototypes to library staff; revise based on feedback	Prototype screens, laptop	Proponents
March 21 - 31, 2026	Conduct system testing	Debugging, functionality tests	Test scripts, laptop	Proponents
April 1 - 10, 2026 (Best-case scenario) April 20 - 24, 2026 (Worst-case scenario)	Prepare final output	Final deployment; prepare documentation and presentations	Documentation files, projector	Proponents

VI. IMPLEMENTATION

DATE	OBJECTIVES	ACTIVITY	MATERIALS NEEDED	PERSON-IN-CHARGE
February 2026	Start project execution	Data gathering, interviews	Notebook, laptop	Proponents
February (Later Weeks) - March (Early Weeks) 2026	System creation	Development, coding, mockups	Computers, software tools	Proponents
March (Later Weeks) 2026	Testing & refinement	System evaluation; fix errors	Test data, laptop	Proponents
April 2026	Finalization	Deployment; submission of documentation	Final system files	Proponents

Prepared by:

JOHN MARK P. LIMSAN
Student, CS Software Engineering 1

RHEA GRACE P. BALATERO
Student, CS Software Engineering 1

Noted by:

CIEMAVIL S. ALCAIN, MIT
Instructor, CS Software Engineering 1

Endorsed by:

LOUELLA MAY V. PLAZA
Student Activity/Scholarship Coordinator

Approved by:

MIGUELITO R. ESPINOSA, MA
Director, Student Services